ARMAN BABAEI

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EDUCATION

Sharif University of Technology September 2024 – Expected August 2026
M.S. in Computer Engineering (Networks)

- Current GPA: 18.50/20; TOEFL: 113/120

Sharif University of Technology September 2019 – July 2024
B.S. in Computer Engineering

- Overall GPA: 19.48/20; Rank: 15/190

HONORS

2026 Ranked **#7** in Iran's National Doctoral Entrance Exam (Networks and Systems subgroup)
2024 Ranked **#4** in Iran's Master University Entrance Exam (Systems and Networks subgroup)
2019 Ranked top **0.01%** in Iran's National University Entrance Exam
2018 **Bronze medalist**, Iran National Olympiad in Informatics (INOI)

RESEARCH EXPERIENCE

Sharif University of Technology August 2025 – Present
L4S/AQM Prototyping for Volumetric Video Streaming *Tehran, Iran*

- Prototyping L4S-style low-latency AQM with DualPI2 in P4_16 (BMv2 `v1model`), including per-egress-port state and ingress-to-egress traffic classification; comparing behavior with Linux DualPI2 and exploring queueing mechanisms for lower latency and better QoS.

Stony Brook University October 2024 – Present
Adaptive Leader Election for PBFT *Remote*

- Designing a modular PBFT-style system with separate core, view-change, and leader-election components; developing an adaptive loop that uses Byzantine-robust aggregation of node-reported progress to adjust leader-selection decisions.

EPFL (École Polytechnique Fédérale de Lausanne) July 2023 – September 2023
Fairness in Decentralized Storage Systems *Lausanne, Switzerland*

- Worked in Bryan Ford's DEDIS lab, supervised by Vero Estrada-Galiñanes. Developed a mathematical model for fairness in Swarm, extended a Go simulator with churn and heterogeneous node behaviors, and designed and evaluated a network-shuffling mechanism that improved fairness. (arXiv)

ISTA (Institute of Science and Technology Austria) July 2022 – February 2023
Parallel Execution in Blockchain Protocols *Vienna, Austria*

- Worked in the SPiDerS lab under Lefteris Kokoris-Kogias and Marios Kogias. Reproduced prior results, built the experimental test bench, analyzed commercial-blockchain workloads, and contributed implementation and algorithmic work for parallel smart-contract execution. (arXiv)

Sharif University of Technology May 2021 – July 2021
Distributed Shortest Path in the CONGEST Model *Tehran, Iran*

- Studied upper and lower bounds for SSSP/MSSP in the CONGEST model and explored approaches for lowering the upper bound for single-source shortest paths.

PUBLICATIONS

Pythia: Supercharging Parallel Smart Contract Execution to Guide Stragglers and Full Nodes to Safety SIB – 2024
Altruism, reciprocity, and tokens to reward forwarding data: Is that fair? ICBC – 2024

WORK EXPERIENCE

Tapsi September 2024 – Present
Proficient Platform Engineer *Tehran, Iran*

- **Apr'26 – Present, Platform Engineering:** Develop and roll out safer Kubernetes delivery workflows using Helm, Flagger, and Istio; improve deployment safety, observability, and developer tooling for production microservices.
- **Sep'25 – Apr'26, SRE:** Operated ride-hailing microservices through monitoring, incident analysis, on-call response, and reliability work across services and deployment infrastructure.
- **Sep'24 – Sep'25, Backend Engineering:** Built and operated passenger–driver matching features; extracted and rewrote Node.js services in Go, including services handling roughly 1k requests/second.

Undisclosed Hedge Fund September 2023 – February 2024
Full-stack Developer *Remote*

- Developed a Python/Django job-scheduler web service for intra-company tasks and its REST APIs.

Mahsan Co. September 2020 – July 2022
C++/Python Developer *Tehran, Iran*

- Developed a Python DSL for end-to-end testing and parallel network-relaying software in C++; also served as Scrum Master for a team of five.

TEACHING ASSISTANT

Operating Systems	F'23	Designed course assignments
Game Theory	S'23	Designed and evaluated practical assignments
Advanced Programming	S'22, S'23	Designed and coordinated the course project
Fundamentals of Programming	F'22	Designed and coordinated the course project
Artificial Intelligence	S'22, F'22	Helped design and grade assignments
Data Structures and Algorithms	S'22	Designed midterm and final exams
Logic Design	S'22	Designed and examined a semester assignment